Constant Proportion Portfolio Insurance (CPPI): Profitability and Risk Analysis under Diffusion and Jump Models

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1 Introduction

Portfolio insurance strategies aim to guarantee a minimum capital at a fixed horizon while preserving exposure to a risky asset in order to capture upside. Two standard constructions are commonly used. *Dynamic* insurance, exemplified by Constant Proportion Portfolio Insurance (CPPI), reallocates wealth between a risk-free asset and a risky asset according to the distance from a predefined floor. *Static* insurance, exemplified by Option-Based Portfolio Insurance (OBPI), secures the floor by investing in a zero-coupon bond and uses the remaining budget to purchase options.

CPPI is attractive in practice because it is simple to implement and responds mechanically to market moves: risky exposure increases after gains and decreases after losses. Its main weakness is *gap risk*: a sufficiently large adverse move between two rebalancing dates may push wealth below the floor, so that the terminal guarantee is no longer met. This risk disappears in the idealized continuous-time limit when price paths are continuous, but it persists in practice due to discrete rebalancing and becomes structural when the underlying price can jump.

This report provides a quantitative and theoretical analysis of CPPI profitability and risk under two market settings: (i) a continuous diffusion model (GBM/Black–Scholes dynamics under the historical measure) and (ii) a Merton jump-diffusion model. We implement a discrete-time CPPI engine with periodic rebalancing and assess risk through the probability and severity of terminal shortfall below the guarantee, complemented by tail measures of the shortfall loss distribution. The numerical study highlights a clear regime difference: under diffusion dynamics, shortfall is primarily a discretization effect, whereas under jump dynamics the floor can be breached with material probability even under frequent monitoring. Finally, we investigate a practical risk-control variant that caps leverage and we benchmark CPPI against OBPI, which delivers a hard terminal floor by construction. The results are interpreted using theoretical jump-risk formulas that link CPPI failure to large negative jumps.

2 CPPI strategy: definition and key mechanisms

2.1 Market and contract

We consider a market with two assets:

- a risk-free money-market account accruing at rate r, and a zero-coupon bond paying N at maturity T. Throughout the report, the floor is defined as the time-t value of this bond: $B_t = Ne^{-r(T-t)}$.
- a risky asset S_t.

The CPPI product starts with initial wealth V_0. In this study we take $V_0 = N$ to simplify interpretation.

2.2 Floor and cushion

At time t, the *floor* is defined as the present value of the guarantee:

$$B_t = Ne^{-r(T-t)}. \tag{1}$$

The *cushion* is

$$C_t = (V_t - B_t)^+. \tag{2}$$

The cushion measures the distance to the floor; it is the budget that CPPI is willing to "risk".

2.3 Allocation rule and leverage

Let $m > 0$ be the *multiple*. At rebalancing dates, CPPI sets the amount invested in the risky asset to

$$E_t = mC_t, \tag{3}$$

and the remaining amount is invested in the risk-free asset:

$$M_t = V_t - E_t. \tag{4}$$

When $m > 1$, E_t can exceed V_t, which corresponds to borrowing (i.e., $M_t < 0$). Hence CPPI is a leveraged, trend-following strategy: after market increases, the cushion grows and risky exposure increases; after market decreases, the cushion shrinks and risky exposure is reduced ("monetarization effect").

2.4 Failure event in discrete time

In practice rebalancing is discrete. A key failure mechanism is: if at a rebalancing date t_k, $V_{t_k} < B_{t_k}$, then the floor is breached (CPPI "fails"). From that date onward we lock the portfolio into the risk-free asset (no further risky exposure), so terminal wealth can remain below N. This is the operational definition used in our implementation.

Remark. In the discrete-time setting, the condition $V_{t_k} < B_{t_k}$ is checked only on rebalancing dates. As a consequence, a floor breach occurring strictly between two dates may only be detected at the next observation time. This is consistent with a trading constraint (positions are adjusted only at rebalancing times), and it explains why the estimated failure probability depends on the monitoring frequency.

3 Theoretical insights: diffusion versus jumps

3.1 Continuous-path model (GBM / Black–Scholes under $\mathbb{P}$)

We assume

$$dB_t = rB_t\,dt, \qquad dS_t = \mu S_t\,dt + \sigma S_t\,dW_t, \tag{5}$$

under the historical measure $\mathbb{P}$. In a continuous-time CPPI idealization with continuous rebalancing, the strategy enforces $E_t = mC_t$ at all times and is self-financing. The cushion dynamics becomes a geometric Brownian motion:

$$dC_t = \hat{\mu} C_t\,dt + \hat{\sigma} C_t\,dW_t, \qquad \hat{\mu} = r + m(\mu - r),\ \hat{\sigma} = m\sigma, \tag{6}$$

implying $C_t > 0$ almost surely for all t, hence the CPPI cannot fail in this continuous-path / continuous-rebalancing idealization. In other words, *gap risk disappears when price paths are continuous and rebalancing is continuous.* Any shortfall observed numerically under GBM is therefore induced by *discrete* rebalancing.

3.2 Jump model and gap risk

To reintroduce gap risk even under frequent monitoring, the risky asset must allow downward jumps. We consider a Merton jump-diffusion:

$$\frac{dS_t}{S_{t^-}} = \mu\,dt + \sigma\,dW_t + (e^Y - 1)\,dN_t, \tag{7}$$

where N_t is a Poisson process of intensity λ, and jump sizes $Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$ are i.i.d. In a jump framework, CPPI can fail because a single large negative jump can push wealth below the floor before the next rebalance.

A useful theoretical result links failure to jump size. Under continuous monitoring and the CPPI rule $E_{t^-} = mC_{t^-}$, a sufficient condition for the cushion to become negative is the occurrence of a jump in the risky return below $-1/m$. For the Merton model, the jump return is $J = e^Y - 1$. Thus the "fatal" condition becomes

$$e^Y - 1 \le -\frac{1}{m} \quad \Longleftrightarrow \quad Y \le \ln\Big(1 - \frac{1}{m}\Big). \tag{8}$$

If ν denotes the Lévy measure of jumps, the probability that at least one jump with size in $(-\infty, -1/m]$ occurs on $[0, T]$ is

$$\mathbb{P}(\text{CPPI fails}) = 1 - \exp\big(-T\,\nu\big((-\infty, -1/m]\big)\big)\,. \tag{9}$$

In the Merton model, such jumps arrive according to a Poisson process. This yields the closed-form approximation

$$\boxed{\mathbb{P}(\text{failure}) \approx 1 - \exp\left(-T\lambda\,\Phi\left(\frac{\ln(1-1/m)-\mu_J}{\sigma_J}\right)\right),} \tag{10}$$

where Φ is the standard normal cdf. This formula provides a direct link between leverage m and gap risk.

4 Numerical methodology

4.1 Global parameters

Table 1 summarizes the contract and market parameters used throughout the notebook.

Table 1: Baseline parameters used in simulations.

Horizon	$T = 5$ years
Guarantee / initial wealth	$N = V_0 = 100$
Risk-free rate	$r = 0.05$ (continuous compounding)
GBM drift / volatility	$\mu = 0.08,\ \sigma = 0.20$
Merton jump intensity	$\lambda = 1.0$ jump/year
Jump log-size	$Y \sim \mathcal{N}(\mu_J = -0.12,\ \sigma_J = 0.18)$
Time grid	252 steps/year ($dt = 1/252$), total steps = 1260
Monte-Carlo size	20,000 paths
Multiples	$m \in \{1, 2, 3, 4, 5, 6\}$
Rebalancing	Daily (1), Weekly (5), Monthly (21) base steps
Transaction costs / borrowing spread	set to 0 in the baseline runs
Leverage-cap variant	$E_t \le L_{\max} V_t, \quad L_{\max} = 1.5$

4.2 Simulation of market returns

The CPPI engine only requires successive *gross returns* $G_{k+1} = S_{k+1}/S_k$. For GBM, log-returns are simulated exactly:

$$\ln G = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\sqrt{dt}\, Z.$$

For the Merton model, we use the standard discretization of log-returns:

$$\ln G = \left(\mu - \frac{1}{2}\sigma^2 - \lambda\kappa\right) dt + \sigma\sqrt{dt}\, Z + \sum_{i=1}^{N} Y_i,$$

with $N \sim \text{Poisson}(\lambda dt)$, $Y_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$, and $\kappa = \mathbb{E}[e^Y - 1]$ (compensator).

4.3 Discrete CPPI engine

Rebalancing occurs every q base steps (daily/weekly/monthly). Between rebalancing dates, the risky sleeve evolves multiplicatively $E \leftarrow E \cdot G$ and cash accrues at e^{rdt} (optionally with a borrowing spread, not activated here). At rebalancing dates, the floor B_t is computed, then the

allocation is reset to $E = m(V - B)^+$. A path is declared failed if $V < B$ at a rebalancing date; thereafter, the portfolio stays fully in cash.

4.4 Risk and performance metrics

We report:

- $\mathbb{E}[V_T]$, median, and low quantiles (5% and 1%).
- Shortfall probability $\mathbb{P}(V_T < N)$ and expected shortfall amount $\mathbb{E}[(N - V_T)^+]$.
- Tail risk on the shortfall loss $L = (N - V_T)^+$: $\mathrm{VaR}_{99}(L)$ and $\mathrm{ES}_{99}(L)$.
- Path-dependent metrics (for selected runs): maximum drawdown (mean and 95% quantile), and turnover (proxy for trading intensity).

5 Core results: $\mathbb{P}(V_T < N)$ and profitability

5.1 Shortfall probability versus multiple m

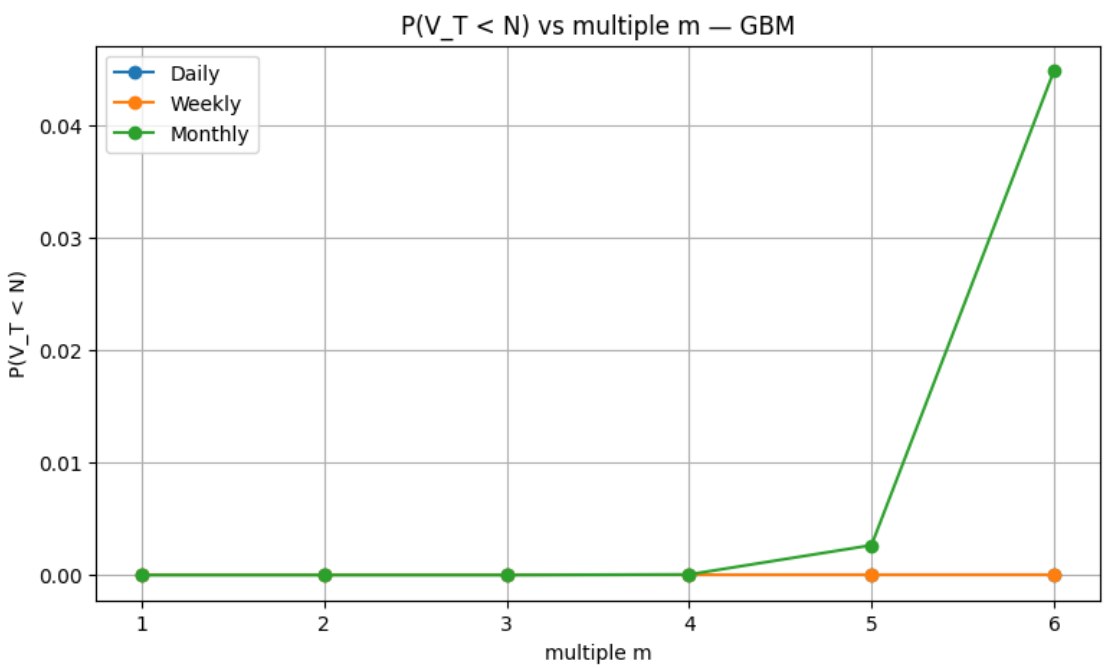

Figure 1: Estimated $\mathbb{P}(V_T < N)$ versus the multiple m under GBM for different rebalancing frequencies (daily/weekly/monthly).

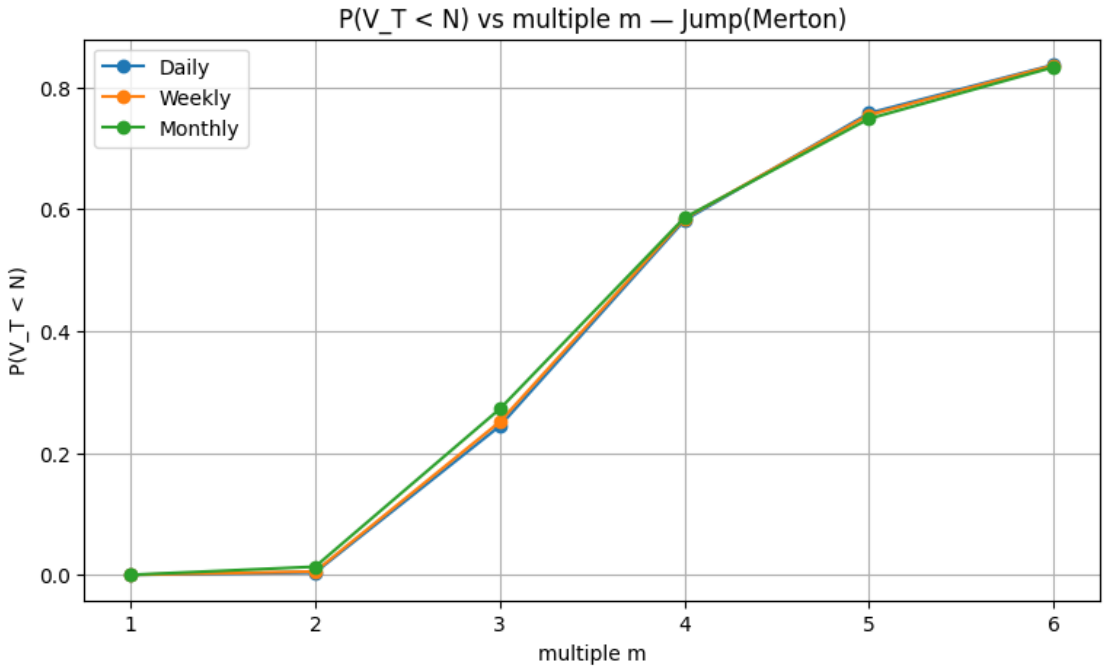

Figure 2: Estimated $\mathbb{P}(V_T < N)$ versus the multiple m under the Merton jump model for different rebalancing frequencies (daily/weekly/monthly).

GBM. Under GBM, the estimated shortfall probability is essentially zero for daily and weekly rebalancing over 20,000 paths (for these configurations, "zero" means $< 1/20{,}000 \approx 5 \times 10^{-5}$). Monthly rebalancing produces a visible shortfall probability for high leverage (notably at $m = 6$, around a few percent). This is consistent with the theory: with continuous paths, CPPI failure is induced mainly by discrete rebalancing (gap risk is a discretization effect).

Jump model. Under Merton jumps, shortfall probability increases very sharply with m and is only weakly affected by rebalancing frequency. This suggests a dominant mechanism: *a single large negative jump* can trigger failure, and increasing rebalancing frequency does not eliminate this risk. For example (daily rebalancing, selected values from the summary table):

$$\mathbb{P}(V_T < N) \approx 0.0039 \ (m = 2), \quad 0.581 \ (m = 4), \quad 0.837 \ (m = 6).$$

This behavior matches the theoretical threshold (8) and probability approximation (10).

5.2 Heatmap summary

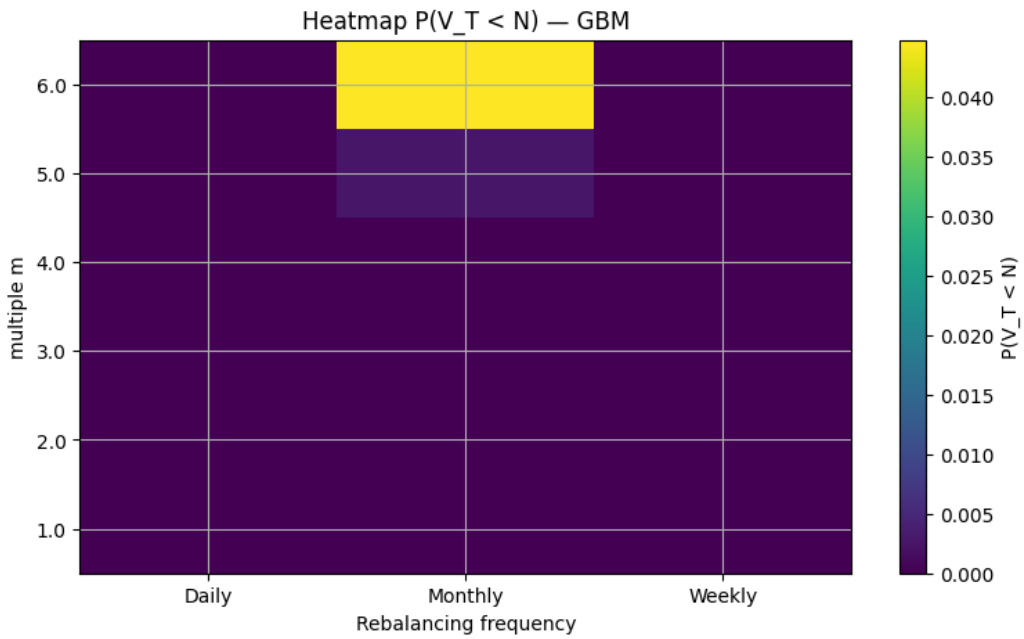

Figure 3: Heatmap of $\mathbb{P}(V_T < N)$ across m and rebalancing frequency under GBM.

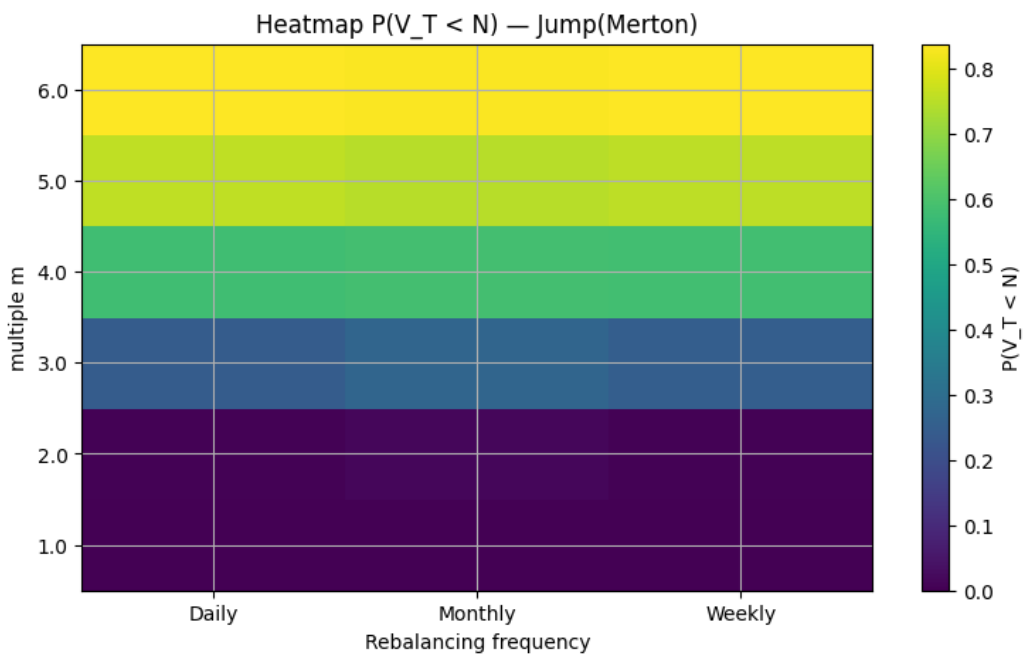

Figure 4: Heatmap of $\mathbb{P}(V_T < N)$ across m and rebalancing frequency under the jump model.

The heatmaps compress the joint sensitivity. Under GBM, risk is concentrated in the low-frequency / high-m corner. Under jumps, m dominates: increasing m dramatically increases the failure probability, while rebalancing frequency has a much smaller effect.

6 Distributional analysis of terminal wealth

A shortfall probability alone does not describe the full risk profile. We therefore compare the terminal distribution V_T under GBM versus the jump model.

6.1 Histograms and empirical CDFs

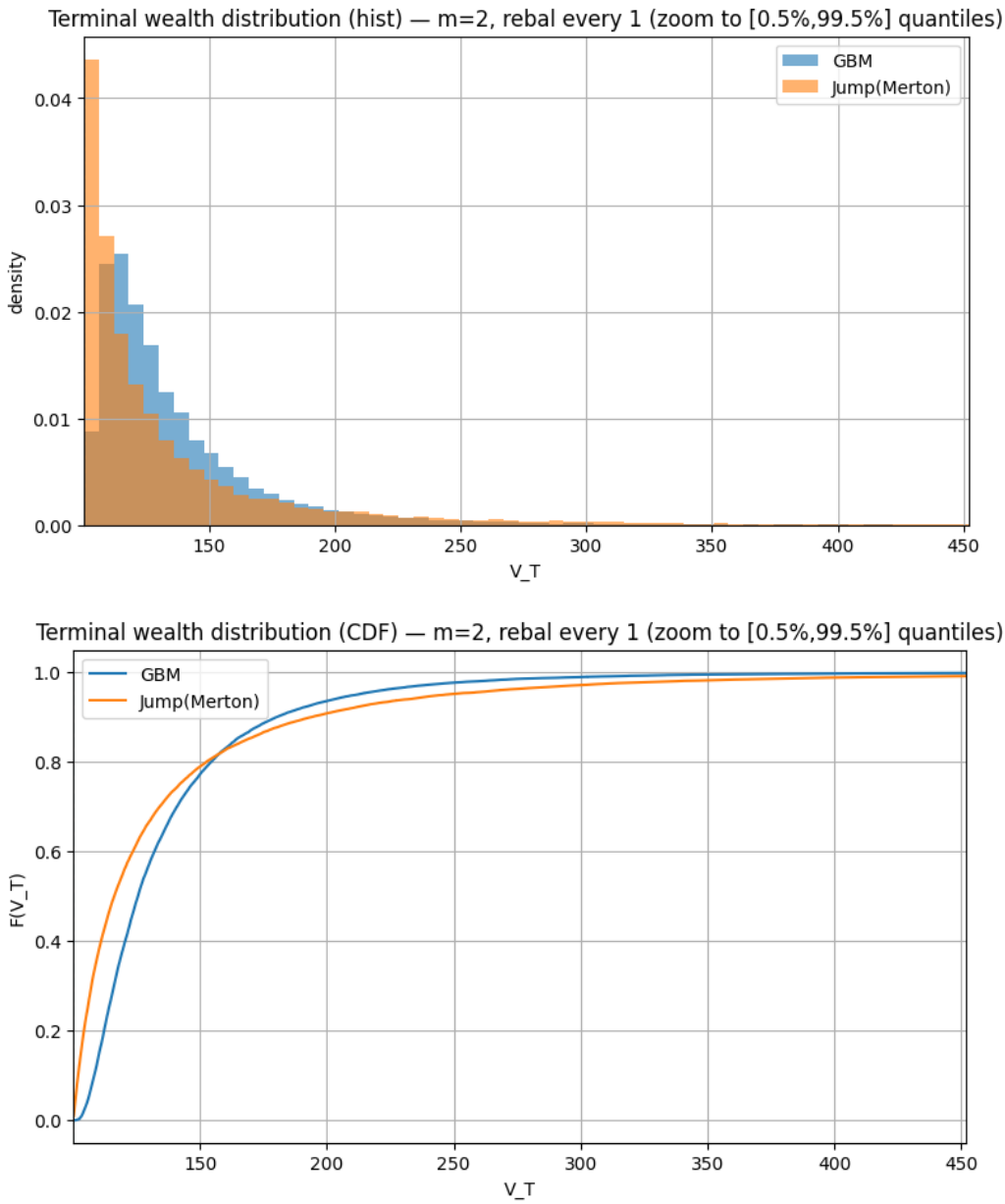

Figure 5: Terminal wealth distribution for $m = 2$ (daily rebalancing): histogram (top) and empirical CDF (bottom), comparing GBM and the jump model.

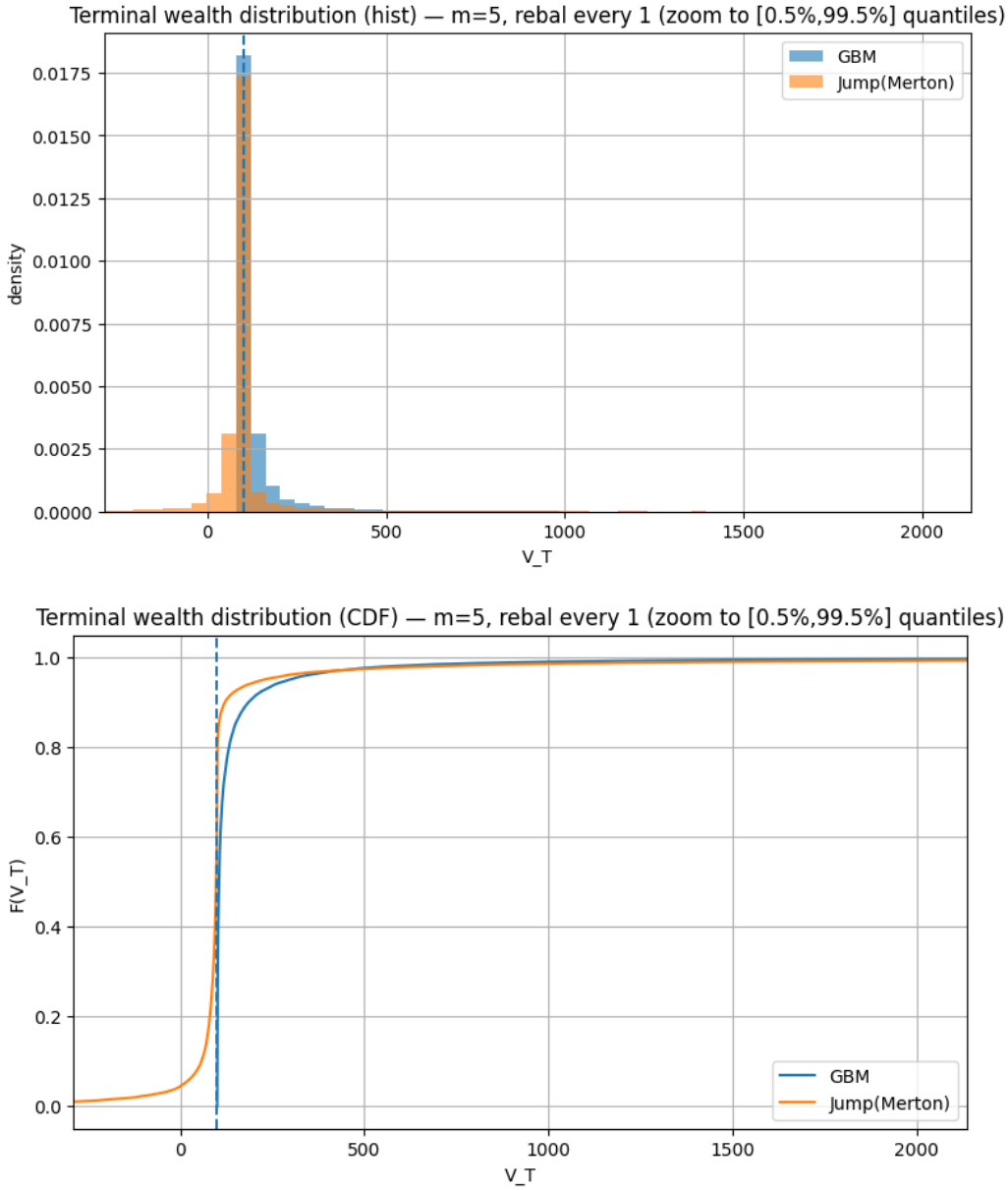

Figure 6: Terminal wealth distribution for $m = 5$ (daily rebalancing): histogram (top) and empirical CDF (bottom), comparing GBM and the jump model.

For $m = 2$, both models exhibit a distribution mostly above the guarantee, but the jump model shows more mass in the left tail. For $m = 5$, differences become much more pronounced: the jump model generates substantial mass below N and far heavier tails. These plots illustrate two important mechanisms: (i) *gap risk*: a jump can cause an abrupt floor breach; (ii) *monetarization*: after a loss, CPPI reduces risky exposure, which limits recovery in some scenarios.

7 Leverage-cap variant: controlling tail risk

A practical way to limit extreme outcomes under leverage is to cap the risky exposure:

$$E_t \leq L_{\max} V_t, \tag{11}$$

with $L_{\max} = 1.5$. This limits borrowing and reduces sensitivity to large downward jumps.

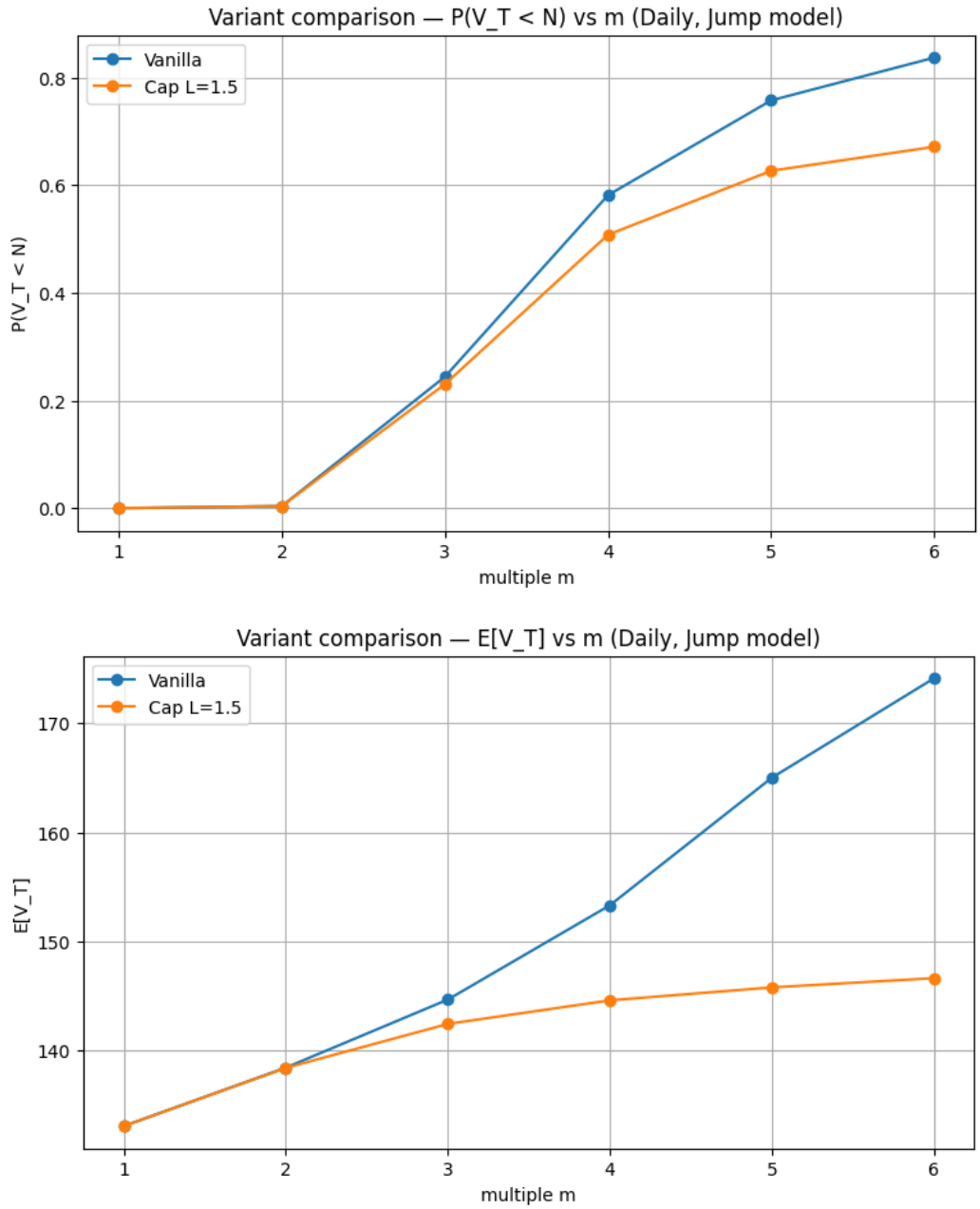

Figure 7: Jump model, daily rebalancing: vanilla CPPI versus leverage-cap CPPI ($L_{\max} = 1.5$). Top: shortfall probability. Bottom: mean terminal wealth.

The cap reduces shortfall risk, especially at high m, at the cost of lower upside. For example (daily, jump model):

$$\mathbb{P}(V_T < N): \ 0.837 \to 0.671 \text{ when } m = 6, \qquad \mathbb{E}[(N - V_T)^+]: \ 79.86 \to 6.23.$$

This illustrates a clear risk–return trade-off: limiting leverage materially improves tail metrics, and also reduces turnover (because the strategy cannot scale risky exposure as aggressively).

8 OBPI benchmark and comparison with CPPI

8.1 OBPI definition

Option-Based Portfolio Insurance (OBPI) constructs a hard guarantee by buying a zero-coupon bond and using the remaining budget to purchase call options. At time 0, the bond investment is Ne^{-rT}. The remaining budget is

$$\text{Budget} = V_0 - Ne^{-rT}.$$

We buy α European calls with strike $K = S_0$ (ATM), where

$$\alpha = \frac{V_0 - Ne^{-rT}}{C_0},$$

and C_0 is the time-0 Black–Scholes call price computed using (r, σ).

The quantity C_0 is used only to satisfy the budget constraint at inception, via the usual Black–Scholes pricing formula. In contrast, the Monte-Carlo scenarios for S_T in this report are generated under the historical measure $\mathbb{P}$ (with drift μ). Hence, we set the OBPI allocation using a time-0 price, and we then compare terminal payoffs under $\mathbb{P}$ in order to discuss economic outcomes.

Terminal wealth is

$$V_T^{\text{OBPI}} = N + \alpha(S_T - K)^+.$$

Ignoring counterparty risk, this implies $V_T^{\text{OBPI}} \geq N$ almost surely.

8.2 Quantitative comparison (m=4, daily)

Table 2 reports the comparison for $m = 4$, daily rebalancing, under both market models.

Table 2: CPPI (m=4, daily) versus OBPI (bond + ATM call). Selected metrics.

Model	Strategy	$\mathbb{E}[V_T]$	Median	$Q05$	$\mathbb{P}(V_T < N)$	$\mathbb{E}[(N - V_T)^+]$
GBM	CPPI (m=4, daily)	149.56	110.20	100.56	0.0000	0.0000
GBM	OBPI (bond + ATM call)	141.07	126.04	100.00	0.0000	0.0000
Jump(Merton)	CPPI (m=4, daily)	153.31	99.75	63.99	0.5812	9.9536
Jump(Merton)	OBPI (bond + ATM call)	147.09	118.98	100.00	0.0000	0.0000

When $\mathbb{P}(V_T < N)$ is reported as 0.0000, no shortfall event was observed among 20,000 simulated paths, so the empirical estimate is below $1/20{,}000 \approx 5 \times 10^{-5}$.

Under GBM, both strategies remain above the guarantee in this simulation; CPPI exhibits higher mean but lower median than OBPI, reflecting a more skewed distribution with higher upside and more dispersion. Under jumps, CPPI displays substantial mass below N, while OBPI retains a hard floor by construction.

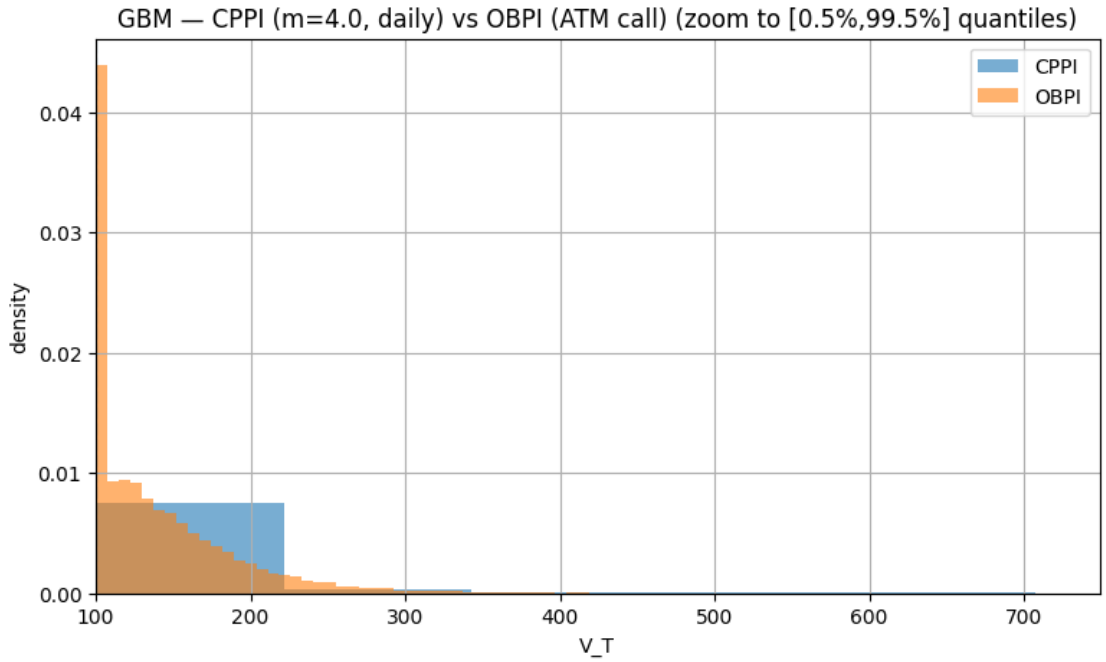

Figure 8: GBM: terminal wealth histogram for CPPI (m=4, daily) versus OBPI (bond + ATM call). Vertical line: guarantee N.

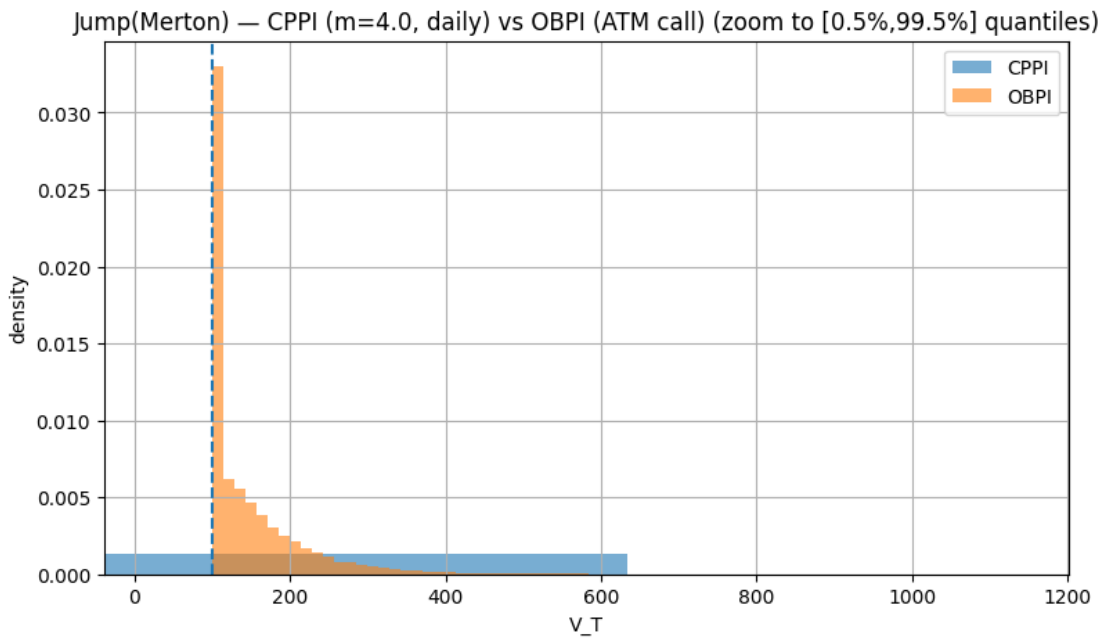

Figure 9: Jump model: terminal wealth histogram for CPPI (m=4, daily) versus OBPI. OBPI has a hard floor $V_T \geq N$, while CPPI shows mass below N due to gap risk.

9 Detailed risk table and interpretation

To provide a compact but informative view of tail risk, Table 3 reports selected cases: GBM vanilla (daily), jump vanilla (daily), and jump with leverage cap $L_{\max} = 1.5$, for $m \in \{2, 4, 6\}$.

Table 3: Summary of selected configurations (daily rebalancing). Metrics are estimated by Monte-Carlo with 20,000 paths.

Case	m	$\mathbb{E}[V_T]$	Median	$Q05$	$\mathbb{P}(V_T < N)$	$\mathrm{VaR}_{99}(L)$	$\mathrm{ES}_{99}(L)$
GBM vanilla	2	137.98	125.42	105.99	0.0000	0.00	0.00
Jump vanilla	2	138.43	116.13	100.89	0.00385	0.00	0.0068
Jump cap $L = 1.5$	2	138.40	116.13	100.89	0.00385	0.00	0.0068
GBM vanilla	4	149.56	110.20	100.56	0.0000	0.00	0.00
Jump vanilla	4	153.31	99.75	63.99	0.5812	139.09	396.73
Jump cap $L = 1.5$	4	144.60	99.996	83.62	0.5077	30.79	38.91
GBM vanilla	6	158.79	101.82	100.02	0.0000	0.00	0.00
Jump vanilla	6	174.11	94.37	-62.31	0.8366	862.24	5033.45
Jump cap $L = 1.5$	6	146.63	98.67	72.05	0.6711	41.85	49.41

Zeros in $\mathbb{P}(V_T < N)$ correspond to "not observed" events over 20,000 paths (i.e. below 5×10^{-5}), rather than a theoretical impossibility.

Two messages emerge clearly:

1. Under GBM, with frequent rebalancing and no transaction costs, downside risk is negligible in this experiment. The strategy mostly reflects a trend-following exposure that increases expected terminal wealth with m.

2. Under jumps, leverage has a dramatic effect on tail risk. For $m = 4$ and $m = 6$, the loss distribution becomes extremely heavy-tailed (very large VaR_{99} and ES_{99}). A leverage cap reduces these tail metrics by an order of magnitude, illustrating the value of practical risk controls.

10 Illustrative wealth paths and mechanism

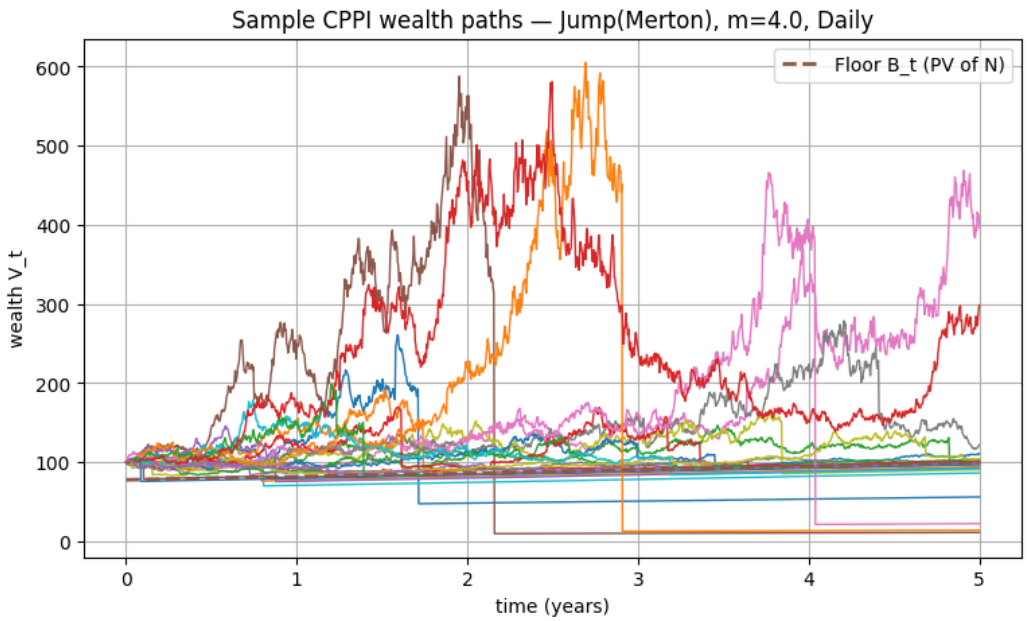

Figure 10: Sample CPPI wealth paths under the jump model (m=4, daily). Dashed line: floor B_t. Several paths experience abrupt drops and then lock into cash below the floor.

The sample paths visualize the key CPPI mechanisms: (i) distance-to-floor tracking (the cushion drives risk-taking), (ii) monetarization after losses (risky weight decreases), and (iii) failure and lock-in after a large downward move. This provides an intuitive explanation for the strong dependence of shortfall risk on m in the jump model.

11 Discussion and limitations

11.1 Why rebalancing matters in GBM but less in jump models

In continuous-path models, gap risk is primarily a discretization phenomenon: more frequent rebalancing keeps the portfolio closer to the ideal continuous-time CPPI rule and reduces floor breaches. Under jumps, however, a single large negative jump can overwhelm the cushion regardless of rebalancing frequency; therefore, increasing monitoring does not eliminate failure risk.

11.2 Role of leverage and practical constraints

Leverage increases upside but also amplifies tail losses. The presence of negative quantiles in some leveraged jump runs reflects the fact that, in the model, borrowing is allowed without margin calls or limited liability. Real implementations would typically include:

- leverage limits (as in our cap variant),
- liquidity and execution constraints (rebalance not always possible at the desired price),
- transaction costs and slippage (which can materially deteriorate performance),

- funding spreads for borrowing (increasing the cost of $M_t < 0$).

These features are important extensions. In our implementation, they are parameterized but set to zero in the baseline runs in order to isolate the gap-risk mechanism.

Trading frictions. In practice, transaction costs and funding spreads may materially reduce CPPI performance, especially when turnover is high (large m, high volatility).
In our code, proportional trading costs and an additional borrowing spread are parameterized; they are set to zero in the baseline runs to isolate gap-risk effects, but they can be activated for robustness checks.

11.3 OBPI versus CPPI

OBPI provides a hard terminal floor N by construction (under the assumption that the bond and the option payoff are delivered). CPPI can provide a *soft* guarantee that holds in continuous-path / continuous-rebalancing idealizations, but can be violated under jumps. Therefore, CPPI and OBPI represent different types of insurance: dynamic versus option-based. In our results, CPPI can have higher mean terminal wealth than OBPI in some settings (especially when m is large), but it can also generate severe shortfalls under jump risk.

12 Conclusion

We studied CPPI with discrete rebalancing under a continuous diffusion model (GBM) and a jump-diffusion model (Merton). The main conclusions are:

- Under GBM, shortfall risk is mainly driven by discrete rebalancing. With frequent rebalancing and zero trading frictions, shortfall is negligible in our experiment.
- Under jumps, CPPI exhibits substantial gap risk. Shortfall probability increases sharply with leverage m, and rebalancing frequency has limited ability to mitigate this risk.
- A leverage cap $E_t \leq L_{\max} V_t$ significantly improves tail risk metrics at the cost of reduced upside and lower trading intensity.
- OBPI provides a hard floor at maturity, highlighting a fundamental difference between dynamic and option-based insurance.

Overall, CPPI is simple and can be attractive for upside participation, but its risk profile is highly sensitive to jump risk and leverage, and practical risk controls are essential.

References

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[2] P. Bertrand and J.-L. Prigent, "Portfolio Insurance Strategies: OBPI versus CPPI,"